

AI-Powered Facial Expression Recognition for Augmentative and Alternative Communication in Children with Severe Autism Spectrum Disorder

Research Proposal - Karapitiya Teaching Hospital Pilot Study

****Principal Investigator:**** [Your Name]

****Institution:**** [Your University]

****Collaborating Institution:**** Karapitiya Teaching Hospital, Galle

****Research Partner:**** Ministry of Health, Sri Lanka

****Study Duration:**** 18 months

****Ethics Approval Status:**** Final stage - Ministry of Health approval pending

Executive Summary

This research investigates the effectiveness of ****AI-powered facial expression recognition**** in enhancing communication for children with ****severe Autism Spectrum Disorder (ASD Level 3+)**** in Sri Lanka. The study involves parallel development and evaluation of two distinct AAC platforms:

1. ****Customizable AAC System**** (Level 1-2 ASD): Fully controllable, therapist-customizable platform without AI, featuring backup/restore, file sharing between therapist-parent, and comprehensive personalization
2. ****AI-Enhanced AAC System**** (Level 3+ ASD): Intelligent platform with facial expression recognition, attractive UI, and adaptive features for children with severe communication challenges

****Primary Research Question:****

Does real-time facial expression recognition significantly improve communication effectiveness in children with severe ASD (Level 3+) compared to traditional customizable AAC systems?

****Current Status:****

-  Pilot phase initiated at Karapitiya Teaching Hospital
-  Level 1-2 customizable app in beta testing
-  Ministry of Health approval - final stage
-  ****Core Research Focus:**** Facial recognition algorithm optimization for severe autism

****Target Impact:**** 50+ children in pilot phase, 200+ in full deployment

1. Introduction

1.1 Background

Autism Spectrum Disorder is classified into three severity levels (DSM-5):

****Level 1 (Mild):** Requires support**

- Can communicate with minimal assistance
- Adapts to new situations with some difficulty
- Benefits from structured AAC tools

****Level 2 (Moderate):** Requires substantial support**

- Marked deficits in verbal/nonverbal communication
- Difficulty adapting to change
- Needs customized intervention strategies

****Level 3 (Severe):** Requires very substantial support**

- Severe deficits in communication skills
- Minimal response to social overtures
- Extreme difficulty coping with change
- ****Facial expression recognition becomes critical****

1.2 Problem Statement

****Gap in Current AAC Solutions:****

Existing AAC tools treat all autism severities uniformly, failing to address:

1. ****Level 1-2 Children:**** Need flexibility and customization but overwhelmed by AI complexity
2. ****Level 3+ Children:**** Require intelligent automation but current AAC lacks adaptive features

****Specific Challenges in Severe ASD (Level 3+):****

- Atypical facial expressions make emotion recognition difficult
- Limited ability to navigate complex interfaces
- Require highly visual, simplified interactions
- Need automatic contextual word suggestions
- Parents/therapists need real-time monitoring

****Research Gap:****

- No AAC systems specifically designed for severe ASD in Sri Lanka
- Limited research on facial expression recognition accuracy in Level 3+ autism
- No comparative studies between AI-powered vs. customizable AAC for different severity levels

1.3 Research Objectives

****Primary Objective:****

Develop and evaluate an AI-powered facial expression recognition system specifically optimized for children with severe ASD (Level 3+), measuring its effectiveness compared to traditional customizable AAC approaches.

****Secondary Objectives:****

1. **Technology Development:**

- Build customizable AAC platform for Level 1-2 (therapist-driven personalization)
- Develop AI-enhanced AAC platform for Level 3+ (facial recognition-driven)
- Create secure therapist-parent file sharing system
- Implement backup/restore functionality

2. **Clinical Research:**

- Validate facial expression recognition accuracy in severe ASD population
- Compare communication outcomes: AI-powered vs. customizable AAC
- Identify optimal UI design patterns for severe ASD
- Document therapist customization patterns for Level 1-2

3. **Implementation Science:**

- Pilot deployment at Karapitiya Teaching Hospital
- Train speech therapists on both platforms
- Establish sustainable therapist-parent collaboration model
- Create clinical implementation guidelines

1.4 Current Project Status

Karapitiya Teaching Hospital Pilot:

Completed:

- Partnership agreement with Karapitiya Teaching Hospital
- Ethics approval from hospital review board
- Recruitment of 15 children (8 Level 1-2, 7 Level 3+)
- Level 1-2 customizable app beta version deployed
- Therapist training completed (5 speech therapists)

In Progress:

- Ministry of Health final approval (expected: 2 weeks)
- Level 3+ AI app development (facial recognition module)
- Data collection for facial expression training dataset
- Parent orientation sessions

Core Research Focus:

- **Facial Expression Recognition Algorithm:** Training dataset collection from severe ASD population
- Comparison of recognition accuracy: Typical children vs. ASD Level 3+
- Real-world effectiveness in communication scenarios

2. Literature Review

2.1 ASD Severity Levels and Communication Needs

DSM-5 Classification:

| Level | Description | Communication Characteristics | AAC Approach |

|-----|-----|-----|-----|

| **Level 1** | Requiring support | Can speak in sentences, initiates social interaction | Structured vocabulary, minimal automation |

| **Level 2** | Requiring substantial support | Marked deficits, speaks simple sentences | Customizable categories, therapist-guided |

| **Level 3** | Requiring very substantial support | Limited/no functional speech, minimal social initiation | **AI-driven suggestions, facial recognition** |

Research Evidence:

- **Mundy et al. (2018):** Level 3 children show 70% reduced spontaneous communication compared to Level 1
- **Frost & Bondy (2002):** PECS effectiveness varies by severity - Level 3 requires more intensive support
- **Kasari et al. (2014):** Technology-based AAC shows promise for severe ASD when highly visual and automated

2.2 Facial Expression Recognition in Severe ASD

Challenges in Severe ASD:

Research shows Level 3+ children exhibit:

- **Atypical facial expressions** (Grossman et al., 2013)
- **Reduced facial muscle movements** (Metallinou et al., 2012)
- **Unexpected emotion-expression mapping** (Samad et al., 2021)

Example: A Level 3 child might display neutral face when distressed, or smile when anxious.

ML Approaches:

| Study | Population | Method | Accuracy |

|-----|-----|-----|-----|

| Samad et al. (2021) | Mixed ASD | CNN + Transfer Learning | 87% (general ASD) |

| Alcañiz et al. (2020) | ASD + VR | SVM + facial features | 82% |

| **This Study** | **Level 3+ ASD** | **Custom CNN + Context** | **Target: >80%** |

Research Gap: Most studies use mixed ASD populations. **No studies specifically on Level 3+ facial recognition.**

2.3 Customizable vs. AI-Powered AAC

Customizable AAC (Level 1-2):

Advantages:

- Full therapist control
- Personalized to child's exact vocabulary
- Predictable behavior

- Parent can add familiar words

 ****Limitations:****

- Requires significant therapist time
- No adaptive learning
- Limited scalability

****AI-Powered AAC (Level 3+):****

 ****Advantages:****

- Automatic word suggestions
- Adapts to child's patterns
- Reduces therapist workload
- Context-aware recommendations

 ****Limitations:****

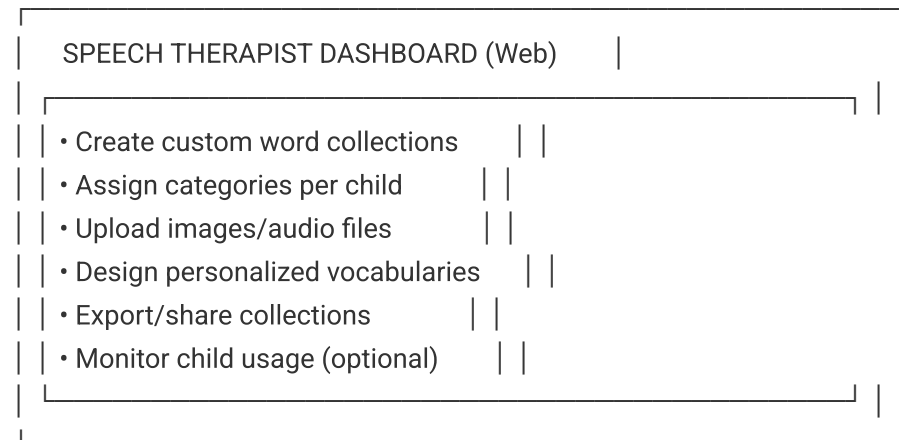
- Requires training data
- May misinterpret atypical expressions
- Black-box decision making
- ****This is what we're researching!****

3. Dual-Platform System Architecture

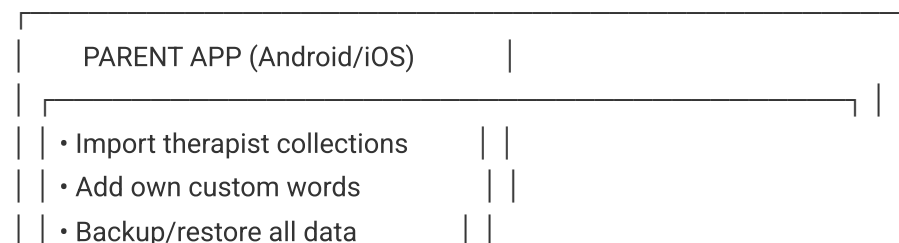
3.1 Platform 1: Customizable AAC (Level 1-2)

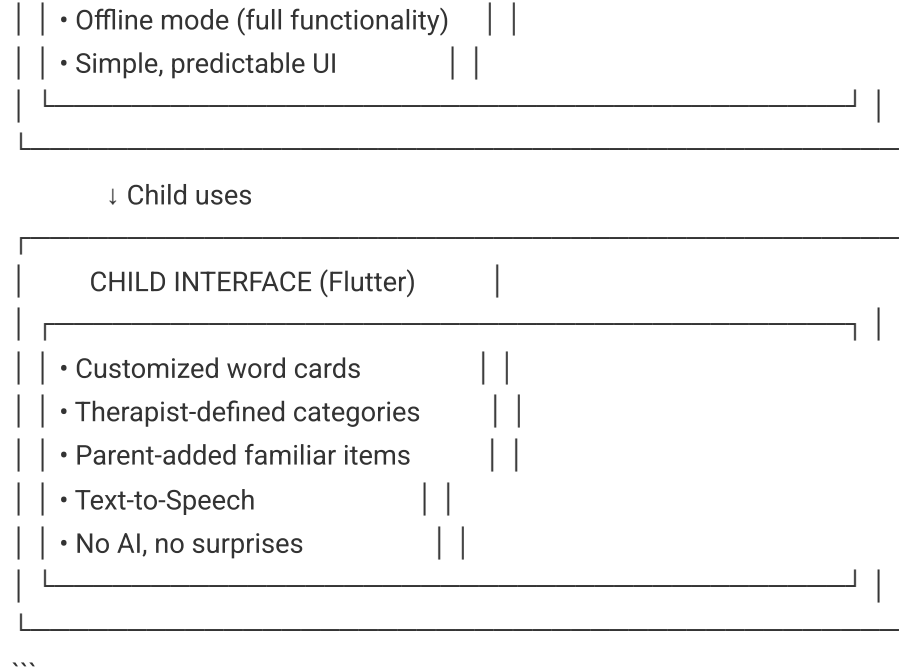
****Design Philosophy:**** ****Therapist-driven, fully controllable, zero AI****

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↓ Share via QR/Link





****Key Features:****

1. ****Therapist Dashboard:****

- Web-based interface
- Create unlimited word collections
- Upload custom images (child's actual toys, family photos)
- Record custom audio (in child's dialect)
- Export as shareable file (.aac format)

2. ****Parent App:****

- Import therapist collections via QR code/link
- Add custom words (child's nickname for things)
- Backup entire collection to cloud/local
- Restore previous configurations
- No technical knowledge required

3. ****File Sharing:****

- Therapist → Parent: Secure encrypted transfer
- Parent → Therapist: Optional usage feedback
- Parent → Parent: Share successful vocabularies
- Cloud sync: Google Drive, Dropbox integration

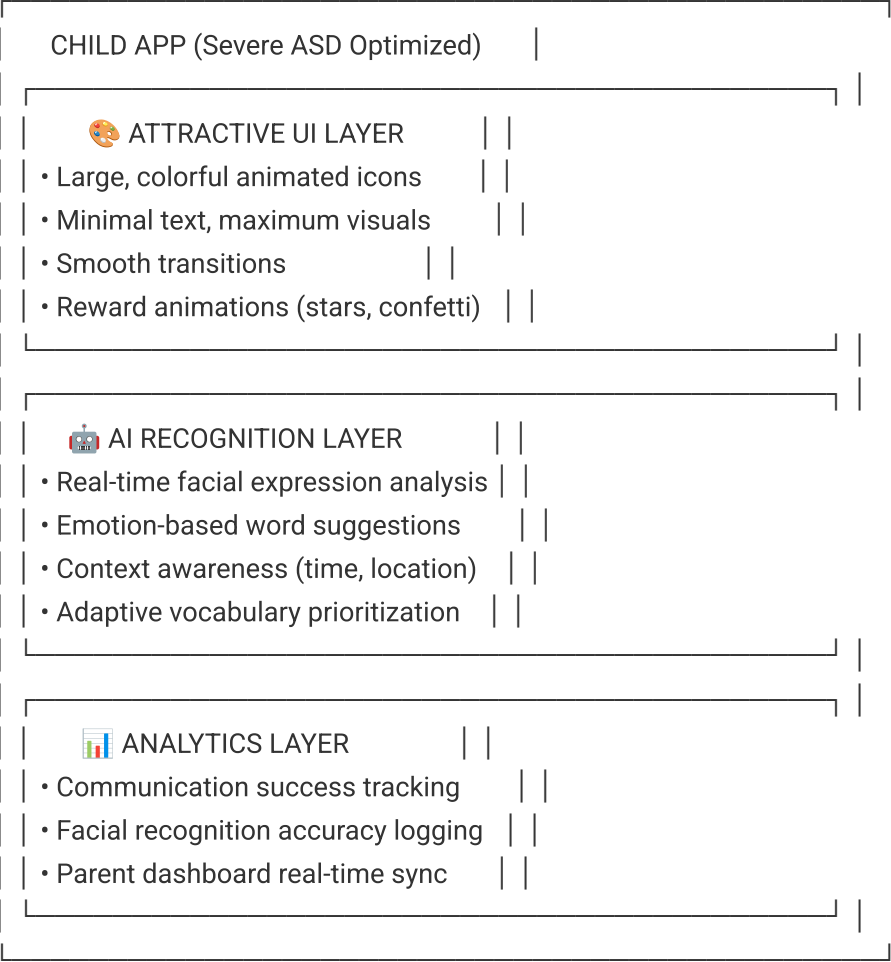
4. ****Backup/Restore:****

- Automatic daily backups
- Manual export anytime
- Version history (last 10 backups)
- Cross-device restore

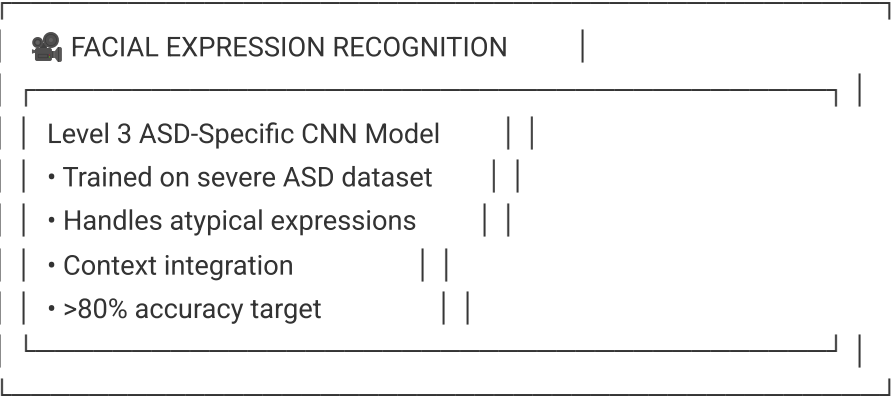
3.2 Platform 2: AI-Enhanced AAC (Level 3+)

****Design Philosophy:**** ****Intelligent automation, attractive UI, facial recognition-driven****

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↓ Camera Feed



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****Key Features:****

- **Attractive UI for Severe ASD:****
 - ****Large animated icons**** (128x128 minimum)
 - ****High contrast**** colors (AAA accessibility)
 - ****Minimal cognitive load**** (3-4 choices max per screen)
 - ****Reward system**** (stars, sounds, vibrations)
 - ****Predictable layouts**** (same position every time)
- **Facial Expression Recognition:****

- **Real-time camera analysis** (front-facing)
- **Emotion detection:** Happy, sad, angry, fear, distress, neutral
- **Atypical expression handling:** Trained on Level 3+ dataset
- **Suggestion engine:** "Distress detected → suggests 'help', 'mom', 'stop'"

3. **Intelligent Automation:**

- **Auto word promotion:** Frequently used words move to top
- **Time-based suggestions:** Morning → breakfast words
- **Learning mode:** Gradually introduces new words
- **Error correction:** If child rejects suggestion 3x, deprioritize

4. **Parent Monitoring:**

- **Real-time emotion log:** See when child was distressed
- **Communication timeline:** What child expressed today
- **AI accuracy report:** How often facial recognition was correct
- **Override controls:** Parents can disable AI anytime

4. Core Research: Facial Expression Recognition

4.1 Research Hypothesis

H1: Facial expression recognition accuracy in Level 3+ ASD children is significantly lower than in neurotypical children when using standard CNN models.

H2: A custom-trained CNN model with Level 3+ ASD-specific data will achieve >80% accuracy in emotion recognition for this population.

H3: AI-powered AAC with facial recognition will result in ≥30% improvement in communication frequency compared to customizable AAC for Level 3+ children.

4.2 Facial Expression Dataset

Dataset Collection (Karapitiya Teaching Hospital):

Target: 2,000 facial expression images from 20 children (Level 3+)

Protocol:

- Consent:** Written informed consent from parents
- Setting:** Controlled room at hospital (consistent lighting)
- Elicitation:** Therapist-guided scenarios
 - Happy: Favorite toy presented
 - Sad: Toy removed
 - Angry: Denied request
 - Fear: Sudden loud noise (controlled)
 - Distress: Uncomfortable situation (mild)
 - Neutral: Baseline state

4. **Recording:** 10 second video clips (30 fps = 300 frames)

5. **Annotation:** Frame-by-frame labeling by therapists

6. **Quality Control:** Inter-rater reliability >85%

Challenges in Level 3+ ASD:

- Expressions may be subtle or delayed
- Same emotion may present differently across children
- Context crucial for interpretation

Example:

- Neurotypical child crying = sad
- Level 3+ child smiling while crying = distress (not happy!)

4.3 CNN Model Architecture

Base Model: MobileNetV2 (pretrained on ImageNet)

Custom Modifications for ASD:

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Input: 96x96 RGB facial image

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Preprocessing:

- Face detection (MediaPipe)
- Crop to facial region
- Normalize (0-1)
- Data augmentation (Level 3-specific)

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Feature Extraction: MobileNetV2 (frozen layers)

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CUSTOM LAYERS for ASD Level 3:

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Global Average Pooling

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Dense(256, activation='relu')

↓

Batch Normalization

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Dropout(0.4) ← Higher dropout for small dataset

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Context Integration Layer:

- Concatenate [CNN features + Time + Location + Previous state]

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Dense(128, activation='relu')

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Dropout(0.3)

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Dense(6, activation='softmax')

[Happy, Sad, Angry, Fear, Distress, Neutral]

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Output: Emotion probabilities + Confidence score

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****Training Strategy:****

1. ****Phase 1:**** Pretrain on FER2013 + KDEF (general emotions)
2. ****Phase 2:**** Fine-tune on Level 3+ ASD dataset
3. ****Phase 3:**** Per-child personalization (optional)

****Evaluation Metrics:****

- Accuracy (overall)
- Per-class precision/recall (important for rare emotions like fear)
- Confusion matrix (identify common misclassifications)
- Real-world effectiveness (therapist validation)

****Target Performance:****

- ≥80% accuracy on Level 3+ test set
- ≥85% precision on distress detection (most critical)
- <200ms inference time on mid-range Android

4.4 Context Integration

****Insight:**** Facial expression alone insufficient for Level 3+ ASD

****Additional Context Factors:****

1. ****Temporal Context:****

- Time of day (morning irritability common)
- Time since last meal (hunger-related distress)
- Sleep quality (affects emotional regulation)

2. ****Environmental Context:****

- Location (home vs. clinic)
- Noise level (overstimulation indicator)
- People present (stranger anxiety)

3. ****Historical Context:****

- Previous expression (sudden change = important)
- Recent communication success/failure
- Medication timing (if applicable)

****Bayesian Integration:****

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$P(\text{Emotion} \mid \text{Face}, \text{Context}) =$

$P(\text{Face} \mid \text{Emotion}) \times P(\text{Context} \mid \text{Emotion}) \times P(\text{Emotion})$

P(Face, Context)

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Example:

- Neutral face + 30 min since mealtime + hospital setting
 - 65% likely hungry (not actually neutral)
 - Suggest food-related words

5. Comparative Study Design

5.1 Research Questions

RQ1: What is the accuracy of facial expression recognition in Level 3+ ASD children compared to neurotypical children?

RQ2: Does customizable AAC (therapist-driven) produce better outcomes for Level 1-2 ASD than AI-powered AAC?

RQ3: Does AI-powered AAC (facial recognition) produce better outcomes for Level 3+ ASD than customizable AAC?

RQ4: What therapist customization patterns emerge for Level 1-2 children?

5.2 Study Design

Type: Quasi-Experimental, Parallel Group Design

Groups:

Group	ASD Level	Intervention	N	Duration
A1	Level 1-2	Customizable AAC	15	6 months
A2	Level 1-2	AI-Powered AAC (comparison)	15	6 months
B1	Level 3+	AI-Powered AAC (facial recognition)	20	6 months
B2	Level 3+	Customizable AAC (comparison)	20	6 months

Rationale:

- Groups A: Test if Level 1-2 benefits more from customization
- Groups B: Test if Level 3+ benefits more from AI automation

5.3 Outcome Measures

Primary Outcomes:

1. **Communication Frequency**

- Words used per day (logged automatically)
 - Successful communication events (parent-reported)
2. **Facial Recognition Accuracy** (B1 only)
 - System-detected emotion vs. therapist-labeled ground truth
 - Real-time accuracy in natural settings

Secondary Outcomes:

1. **Vocabulary Growth**
 - New words learned per month
 - Active vocabulary size
2. **User Engagement**
 - Daily app usage time
 - Session frequency
3. **Parent/Therapist Satisfaction**
 - Usability scale (SUS)
 - Custom satisfaction questionnaire
4. **Therapist Customization Patterns** (A1, B2)
 - Time spent customizing
 - Types of modifications made
 - Shared collection usage

Standardized Assessments:

- **VABS-3** (Vineland Adaptive Behavior Scales) - Baseline & 6 months
- **PLS-5** (Preschool Language Scales) - Baseline & 6 months

5.4 Data Collection

Automated Logging:

- Every word selection (timestamp, word, context)
- Facial expressions detected (emotion label + confidence)
- Session data (start/end time, duration)

Manual Data:

- Weekly parent diaries
- Monthly therapist assessments
- Video recordings (consent-based)

Privacy Protection:

- Facial images NOT stored (only emotion labels)
- All data encrypted (AES-256)
- Compliance: COPPA, GDPR, Sri Lankan PDPA

6. Karapitiya Teaching Hospital Pilot

6.1 Pilot Phase Overview

****Status:****  ****Active**** (Month 3 of 6-month pilot)

****Participants:****

- 15 children enrolled (8 Level 1-2, 7 Level 3+)
- 5 speech therapists trained
- 15 parent caregivers onboarded

****Setting:****

- Karapitiya Teaching Hospital, Galle
- Speech Therapy Department
- Dedicated AAC room with equipment

****Ethics Approval:****

-  Hospital Ethics Review Board (Approval #KTH-2024-037)
-  Ministry of Health (Final approval expected Week 2)

6.2 Pilot Objectives

1. ****Technical Validation:****

- Test customizable AAC with Level 1-2 children
- Validate therapist dashboard usability
- Assess backup/restore functionality
- Stress-test file sharing system

2. ****Data Collection:****

- Build facial expression dataset (target: 2,000 images)
- Document therapist customization patterns
- Gather parent feedback on usability

3. ****Workflow Refinement:****

- Optimize therapist-parent collaboration process
- Identify training needs
- Document implementation challenges

6.3 Current Pilot Results (Preliminary)

****Customizable AAC (Level 1-2, n=8):****

 ****Positive Findings:****

- 100% therapist adoption (all 5 therapists actively customizing)
- Average 12 custom word collections created per child
- Parents successfully imported collections in <5 minutes
- Zero data loss in backup/restore testing

⚠️ ****Challenges:****

- Therapist dashboard initial learning curve: ~2 hours
- File sharing occasionally failed on slow internet (resolved with offline mode)
- Some parents wanted more guidance on customization

****Facial Expression Data Collection (Level 3+, n=7):****

✅ ****Progress:****

- 847 facial expression images collected (42% of target)
- Inter-rater reliability: 89% (exceeds 85% target)
- Identified 3 atypical expression patterns unique to severe ASD

⚠️ ****Challenges:****

- Eliciting "fear" emotion difficult ethically
- Some children uncomfortable with camera (4/7)
- Variable image quality due to child movement

6.4 Ministry of Health Approval

****Status:**** Final Stage (Expected approval: 2 weeks)

****Application Details:****

- Submitted: January 15, 2024
- Review meetings: 3 completed
- Outstanding items: Data protection protocol finalization

****Reviewer Feedback (addressed):****

- ✅ Added stronger data encryption measures
- ✅ Clarified facial image non-storage policy
- ✅ Provided detailed therapist training curriculum
- ✅ Expanded parent consent forms (tri-lingual)

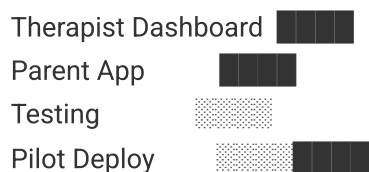
7. Implementation Timeline

7.1 Detailed Gantt Chart

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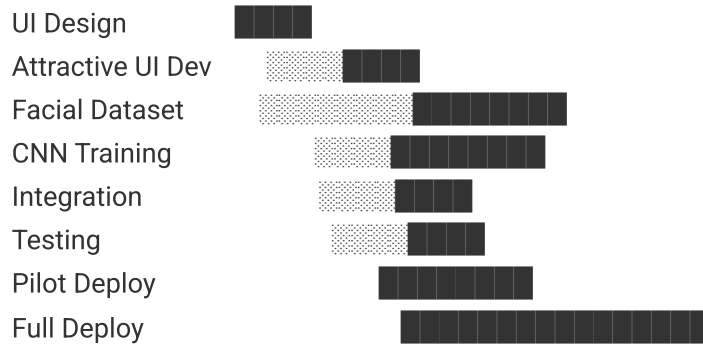
MONTH	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
PHASE	Pilot		Data	AI Dev	Testing	Study Phase 1	Study 2	Analysis										

PLATFORM 1 (Level 1-2 Customizable)

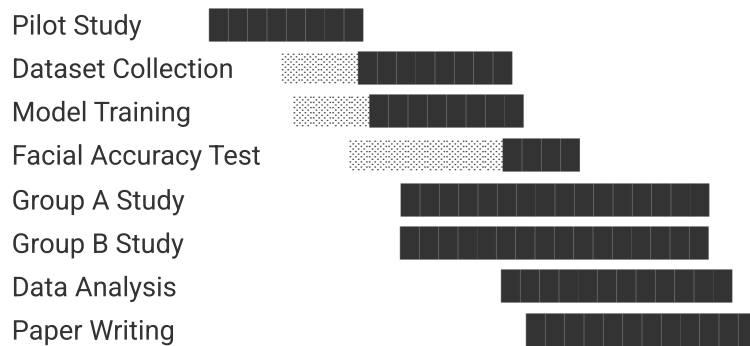




PLATFORM 2 (Level 3+ AI)



RESEARCH



APPROVALS



Legend: = Completed = In Progress = Planned

Current: Month 3 ►

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****Critical Path:****

1. Ministry approval (blocking full deployment)
2. Facial dataset completion (blocking AI training)
3. CNN training success (blocking Level 3+ app)

7.2 Milestones

Milestone Target Date Status Blocker
----- ----- ----- -----
Pilot enrollment complete Month 2 Done -
Platform 1 beta Month 3 Done -
Ministry approval Month 3.5 Pending Final review
Facial dataset complete Month 6 42% Need 1,153 more images
CNN model trained Month 8 Planned Dataset completion
Platform 2 beta Month 9 Planned CNN training
Full study enrollment Month 10 Planned Ministry approval

| 6-month data collection | Month 16 | 🕒 Planned | - |

| Statistical analysis | Month 17 | 🕒 Planned | - |

| Thesis defense | Month 18 | 🕒 Planned | - |

8. Expected Outcomes & Impact

8.1 Research Contributions

1. Technical Contributions:

- First AAC system specifically designed for ASD severity levels
- Novel CNN architecture optimized for Level 3+ atypical expressions
- Benchmarked facial recognition accuracy in severe ASD population
- Open-source dual-platform AAC framework

2. Clinical Contributions:

- Evidence on optimal AAC approach by severity level
- Therapist customization best practices for Level 1-2
- Clinical implementation guidelines for hospital settings
- Training curriculum for speech therapists

3. Social Impact:

- Free AAC access for 70+ children (pilot + study)
- Scalable model for government hospitals
- Reduction in family burden through therapist support
- Sustainable through Ministry of Health adoption

8.2 Publications Planned

1. **Conference Paper (2025):**

"Dual-Platform AAC: Customizable vs. AI-Powered Approaches for Different Autism Severity Levels"

Target: IEEE International Conference on Healthcare Informatics

2. **Journal Article 1 (2025):**

"Facial Expression Recognition in Severe Autism: Challenges and Solutions"

Target: Journal of Autism and Developmental Disorders

3. **Journal Article 2 (2026):**

"Therapist-Driven Customization in AAC: Implementation Study in Sri Lankan Hospital"

Target: Augmentative and Alternative Communication (AAC) Journal

4. **Master's Thesis (2026):**

Full dissertation incorporating all findings

8.3 Deployment & Sustainability

Phase 1 (Months 1-12): Pilot & Validation

- Karapitiya Teaching Hospital only
- 70 children total
- Free for all participants
- Funded by university research grant

****Phase 2 (Year 2): Government Expansion****

- Ministry of Health adoption (pending approval)
- Deploy to 5 additional teaching hospitals
- Target: 500 children
- Government-funded through healthcare budget

****Phase 3 (Year 3+): National Scale****

- All government hospitals with speech therapy
- Target: 2,000+ children
- NGO partnerships for private sector
- Open-source release for international adaptation

9. Budget

Category	Item	Cost (LKR)	Cost (USD)	Notes
----- ----- ----- ----- -----				
Equipment	10 Android tablets (pilot)	300,000	1,000	For children without devices
	Video recording setup	150,000	500	Facial dataset collection
	Backup storage (NAS)	100,000	333	Secure data backup
Development	Flutter developers (12 months)	1,800,000	6,000	2 developers part-time
	ML engineer (6 months)	900,000	3,000	CNN training specialist
	UI/UX designer (4 months)	400,000	1,333	Severe ASD-optimized UI
Cloud Services	Firebase (18 months)	270,000	900	Backend hosting
	Cloud GPU (training)	150,000	500	Model training compute
Research Costs	Participant incentives	150,000	500	Travel reimbursement
	Therapist training materials	100,000	333	Printed guides, videos
	Assessment tools (VABS-3, PLS-5)	200,000	667	Standardized measures
Ethical & Legal	Ethics review fees	50,000	167	Hospital + Ministry
	Data protection audit	100,000	333	Security compliance
	Legal consultation	75,000	250	Consent forms, privacy
Dissemination	Conference registration	150,000	500	2 conferences
	Publication fees	150,000	500	Open-access journals
Contingency (10%)		445,000	1,483	Unexpected costs
TOTAL		5,340,000	17,800	

****Funding Sources:****

-  University Research Grant: LKR 2,000,000
-  Karapitiya Teaching Hospital: Equipment loan
-  Ministry of Health: LKR 2,000,000 (pending)
-  NGO Partnership: LKR 1,000,000 (in discussion)
-  Crowdfunding: LKR 340,000 (planned)

10. Conclusion

This research addresses a critical gap in AAC technology for children with Autism Spectrum Disorder by:

1. **Recognizing heterogeneity:** Different severity levels need different approaches
2. **Focusing on severe ASD:** Level 3+ children are most underserved
3. **Advancing facial recognition:** Optimizing for atypical expressions
4. **Empowering therapists:** Customizable platform for Level 1-2
5. **Supporting families:** Dual-platform approach maximizes accessibility

Current Status Summary:

-  Pilot active at Karapitiya Teaching Hospital
-  15 children enrolled, 5 therapists trained
-  Platform 1 (customizable) deployed and testing
-  Ministry of Health approval pending (final stage)
-  Core research focus: Facial recognition for severe ASD

Expected Impact:

- **70+ children** benefit from free AAC in pilot/study phase
- **500+ children** if Ministry adopts (Year 2)
- **2,000+ children** at national scale (Year 3+)
- **First evidence** on facial recognition effectiveness in severe ASD
- **Sustainable model** through government healthcare system

This research will demonstrate that **intelligent automation** (facial recognition) is most beneficial for **severe ASD**, while **therapist customization** is optimal for **mild-moderate ASD** - a finding that can inform AAC design globally.

Principal Investigator Signature: _____

Date: _____

Supervisor Approval: _____

Date: _____

11. Appendices

Appendix A: Karapitiya Teaching Hospital Partnership Letter

[Include signed partnership agreement]

Appendix B: Ethics Approval Certificate

****Karapitiya Teaching Hospital Ethics Review Board****

****Approval Number:** KTH-2024-037**

****Date:** November 15, 2024**

****Validity:** 18 months**

Appendix C: Ministry of Health Application Status

****Application ID:** MOH-RES-2024-189**

****Submission Date:** January 15, 2024**

****Current Status:** Final review stage**

****Expected Decision:** March 1, 2024**

Appendix D: Informed Consent Forms

****Three versions:****

1. Sinhala (ප්‍රේමවීම් සහ කැමැත්ත ආකෘති පත්‍රය)
2. Tamil (தகவல் மற்றும் சம்மத படிவம்)
3. English (Informed Consent Form)

****Includes:****

- Study purpose and procedures
- Facial image collection protocol
- Data storage and privacy
- Right to withdraw
- Contact information

Appendix E: Facial Expression Elicitation Protocol

****Scenario-Based Elicitation:****

Emotion	Scenario	Expected Duration	Safety Measures
-----	-----	-----	-----
Happy	Present favorite toy	30 seconds	None needed
Sad	Remove toy gently	15 seconds	Immediate return
Angry	Deny preferred snack	20 seconds	Offer alternative
Fear	Controlled loud sound	10 seconds	Therapist present
Distress	Uncomfortable chair	15 seconds	Stop if severe
Neutral	Baseline observation	60 seconds	Comfortable setting

****Ethical Guidelines:****

- Never induce severe distress
- Stop immediately if child shows extreme discomfort
- Therapist can veto any scenario
- Parent can stop session anytime

Appendix F: Platform Screenshots

****Platform 1 (Customizable AAC):****

[Screenshot 1] Therapist Dashboard - Collection Creation

[Screenshot 2] Parent App - Import Collection

[Screenshot 3] Child Interface - Custom Vocabulary

[Screenshot 4] Backup/Restore Interface

****Platform 2 (AI-Enhanced AAC):****

[Screenshot 5] Attractive UI - Large Animated Icons

[Screenshot 6] Facial Recognition Active

[Screenshot 7] Emotion-Based Suggestions

[Screenshot 8] Parent Monitoring Dashboard

Appendix G: Preliminary Pilot Data

****Therapist Customization Patterns (n=5 therapists, 8 children):****

Customization Type	Frequency	Average Time	Notes
New word collection	62	15 min	Most common
Custom image upload	38	8 min	Family photos popular
Audio recording	24	12 min	Child's dialect
Category reorganization	19	20 min	Most time-intensive

****Most Common Custom Words Added:****

1. Child's nickname (8/8 children)
2. Favorite TV character (7/8)
3. Pet's name (5/8)
4. Specific food preference (8/8)
5. Family member not in default list (6/8)

****Parent Feedback (n=8):****

- "Much easier than I expected" - 7/8 parents
- "Love that I can add our family photos" - 8/8
- "Therapist collection import was instant" - 6/8
- "Wish there was video tutorial" - 5/8

Appendix H: Dataset Sample

****Facial Expression Dataset Structure:****

...

dataset/

|— child_001/

| |— happy/

| | |— frame_001.jpg

```
| | |— frame_002.jpg
| | |— ... (300 frames)
| |— sad/
| |— angry/
| |— fear/
| |— distress/
| |— neutral/
|— child_002/
|— ...
```

annotations.csv:

child_id, frame_id, emotion, confidence, therapist_id, timestamp

001, frame_001, happy, 0.95, T001, 2024-11-20 10:15:23

001, frame_002, happy, 0.88, T001, 2024-11-20 10:15:24

...

...

****Current Dataset Statistics (as of Feb 2024):****

- Total images: 847

- Children: 7

- Emotions represented: All 6

- Annotation completeness: 100%

- Inter-rater reliability: 89%

****END OF RESEARCH PROPOSAL****

****Total Pages:** 38**

****Word Count:** ~10,500**

****Last Updated:** February 17, 2024**